

Coin Rush Expansion 01

Animated Characters and Magic Combat

Upgrade the first game with skeletal models, locomotion graphs, montage-like attacks, animation events, staff projectiles, enemy attack timing, audio, particles, and runtime validation.

REQUIRES

The completed Coin Rush first-game milestone or an equivalent scene containing PlayerStart, Enemy, Health, navigation, HUD, registered game scripts, and a valid runtime export.

Animated-combat expansion map

Preserve the working collection game while adding animation and combat in small, reversible stages.

ID	Stage	Deliverable	Exit gate
X0	Baseline lock	Duplicate scene / verify original	Old victory and defeat still pass
X1	Character assets	Player/enemy skeletal models	Models stand upright at correct scale
X2	Locomotion	Idle/Walk/Run controllers	Speed drives smooth movement
X3	Player cast	StaffCast Action Profile	Input plays once and returns
X4	Magic event	StaffRelease + projectile	Projectile spawns on event
X5	Projectile hit	Trigger damage script	Enemy loses health once
X6	Enemy animation	Locomotion + EnemyAttack	AI moves and attacks visibly
X7	Combat timing	EnemyHit event damage	Damage matches contact frame
X8	Feedback	Audio, particles, camera	Cast/hit are readable
X9	Release	Playtest, F7, F8, package	Expansion works standalone

SAFETY RULE

Work in a new scene such as Scenes/CoinRush_Animated.scene. Never trade a proven playable baseline for several unfinished systems.

Import and prepare both characters

Animation tuning cannot repair an incorrect model basis, scale, skeleton, or missing clips.

1

Import models

Place player and enemy FBX/model assets under the project Assets tree.

2

Create scene objects

Drag each model into the expansion scene and enable Skeletal Model in Inspector > Animation.

3

Inspect inventory

Confirm non-zero Clips and Bones; write down clip names and durations.

4

Check orientation

The character should stand upright and face the engine forward direction.

5

Check scale/pivot

Fit the visual model to the controller/agent capsule without extreme object scale.

6

Preview clips

Use Panels > Animation Preview to inspect Idle, Walk, Run, Cast/Attack, Hit, and Death.

Check	Pass	Repair before continuing
Basis	Upright and forward	Correct import/model orientation
Root	In-place locomotion stays centered	Use correct clips or intentional root motion
Loops	Idle/Walk/Run have clean seams	Repair source animation/clip
Skeleton	Stable bone names and hierarchy	Re-export rig consistently
Scale	Capsule and feet align	Correct import/object scale

DO NOT ROTATE AROUND A BAD IMPORT

A visual correction on the object can make animation forward, translation, and local gizmo axes disagree. Fix the model/skeleton basis first when possible.

Combine visual models with gameplay bodies

Keep the existing PlayerStart and Enemy entities because their components, names, teams, Health, and scripts are already wired.

- Assign the animated model asset to the existing gameplay object when the editor supports it; avoid creating a second controller entity with the same role.
- Align the model feet with the bottom of the player capsule and center the hips over the capsule axis.
- For Enemy, keep Nav Agent, Health, collider, team, Behavior Graph, and animated model on the same runtime entity.
- Keep PlayerStart and Enemy names unchanged because scripts locate them by name.
- Use selection outlines and Skeleton view to distinguish model geometry from capsule/agent debug guides.

Recommended ownership

Concern	Owner
Player translation/collision	Player Controller capsule
Player pose	AnimatedModel controller + Action Profiles
Enemy translation/path	Nav Agent / Behavior Graph
Enemy pose	AnimatedModel controller + actions
Damage	Animation-event-driven scripts
Effects	Particle/Audio components or prototypes
ONE ENTITY, ONE STATE	The movement-lock behavior for full-body actions is read from the AnimatedModel on the moving entity. Splitting animation and movement across unrelated entities prevents that integration.

Set up Idle, Walk, and Run

Use the quick locomotion helper first. Move to a State Graph only when jump/fall/hit/death rules need explicit transitions.

- 1

Default clip
Set Idle as the Default Clip; enable Autoplay and Loop.
- 2

Locomotion States
Enable Use Locomotion and assign Idle, Walk, Run.
- 3

Thresholds
Start with Walk At 0.15 and Run At 3.0.
- 4

Player test
The built-in player update writes Speed from movement and sprint values.
- 5

Enemy test
Nav-agent velocity drives the animation Speed value in runtime integration.
- 6

Tune stride
Adjust state/clip speed or gameplay speed until feet no longer slide noticeably.

Speed	State	Expected
0	Idle	Stationary loop
0.15 to <3	Walk	Normal travel
3+	Run	Sprint/fast chase

CONTROLLER
PRIORITY

If State Graph contains states, it takes priority over the simple Use Locomotion helper. Do not tune the helper while an authored graph is active.

Add jump, hit, and death states

The State Graph supports Float, Bool, and Trigger parameters, transitions, priorities, exit time, interruptible blends, optional blend clips, and per-state root motion.

Parameter	Type	Use
Speed	Float	Idle/Walk/Run transitions or blend weight
Grounded	Bool	Jump/fall/land rules
Hit	Trigger	Any State -> HitReaction
Dead	Bool	Any State -> Death
Attack	Trigger	Use only if attack is a graph state

Suggested rules

From	To	Condition	Exit	Priority
Idle	Walk	Speed >= 0.15	0	0
Walk	Idle	Speed < 0.15	0	0
Walk	Run	Speed >= 3.0	0	0
Run	Walk	Speed < 3.0	0	0
Any State	HitReaction	Hit trigger	0	30
Any State	Death	Dead == true	0	100

- Use 0.15-0.25 s fades for locomotion and shorter fades for urgent hit reactions.
- Death should not loop and should have the highest priority.
- Keep controller/navmesh locomotion in-place; enable Root Motion only for authored displacement such as a lunge.
- Use Graph Validation in Edit mode and Transition Debug in Play mode.

Create the StaffCast Action Profile

Action Profiles provide montage-like one-shot playback while the locomotion controller continues underneath.

Field	Committed cast	Moving cast
Name	StaffCast	StaffCastMoving
Clip	Staff cast/attack clip	Same or lighter clip
Mask Root	Full Body / empty	Spine or chest bone
Fade In	0.08 s	0.08 s
Fade Out	0.15 s	0.15 s
Speed	1.0	1.0

1

Action Test

Choose the cast clip, fades, speed, and mask; click Play Action.

2

Choose movement policy

Empty mask locks player/AI movement until completion. A bone mask keeps locomotion active.

3

Save profile

Add Profile, name it StaffCast, and keep the name stable for scripts.

4

Input test

Use one-shot mouse input; never restart the action every frame.

5

Return test

After completion, the underlying locomotion pose should resume cleanly.

RECOMMENDED FIRST PASS

Use a full-body StaffCast so the timing is easy to read. Add a masked moving-cast variant only after the projectile and damage loop work.

Add StaffRelease and a projectile prototype

Spawn gameplay at the animation event, not at the input press. The wind-up is then visible and the release stays aligned when animation speed changes.

1

Scrub the cast

Pause the clip and scrub to the frame where magic leaves the staff.

2

Add Event

Create StaffRelease on the cast clip at the current time.

3

Prototype

Create FireballPrototype outside the playable area with model/primitive, trigger collider, particles, optional audio, and FireballProjectile script.

4

Spawn

StaffCombat listens for StaffRelease and clones FireballPrototype.

5

Transform

Set the clone position in front of the caster and copy the caster rotation.

STAFFCOMBAT CORE

```
class StaffCombat : public engine::Script {
public:
    void OnUpdate(float) override {
        if (WasMouseButtonPressed(0) &&
            !IsAnimationActionPlaying())
            PlayAnimationProfile("StaffCast");

        if (!WasAnimationEvent("StaffRelease")) return;
        auto* me = Transform();
        if (!me) return;
        const glm::vec3 forward =
            me->rotation * glm::vec3(0, 0, -1);
        const auto fireball = SpawnFromObject(
            "FireballPrototype",
            me->position + glm::vec3(0, 1.1f, 0) + forward);
        if (auto* t = TryGet<engine::ecs::Transform>(fireball))
            t->rotation = me->rotation;
    }
};
```


Move, collide, damage, and expire

The projectile clone carries its own script and trigger. It moves along its rotation forward vector and damages Enemy once on trigger entry.

FIREBALLPROJECTILE CORE

```
class FireballProjectile : public engine::Script {
    float life = 0.0f;
public:
    void OnUpdate(float dt) override {
        life += dt;
        auto* t = Transform();
        if (!t) return;
        const float speed = GetFieldFloat("speed", 12.0f);
        const glm::vec3 forward =
            t->rotation * glm::vec3(0, 0, -1);
        t->position += forward * speed * dt;

        const auto enemy = FindObject("Enemy");
        if (enemy != engine::ecs::kNull &&
            WasTriggerEntered(enemy)) {
            if (auto* hp = TryGet<engine::Health>(enemy))
                hp->Damage(GetFieldFloat("damage", 25.0f));
            DestroySelf();
            return;
        }
        if (life >= GetFieldFloat("lifetime", 4.0f))
            DestroySelf();
    }
};
```

Script field	Type	Starter
speed	Float	12.0
damage	Float	25.0
lifetime	Float	4.0

COLLIDER RULE

Use a trigger collider on the projectile. A blocking rigid-body projectile can bounce or push the target before the gameplay event is processed.

Animate the navigating enemy

Configure enemy Idle/Walk/Run exactly like the player, but tune thresholds to the Nav Agent speed rather than keyboard movement.

- Keep AnimatedModel, Nav Agent, Behavior Graph, Health, collider, and team on Enemy.
- Use in-place locomotion because navigation owns translation.
- Preview the enemy attack clip and create full-body EnemyAttack Action Profile.
- Add EnemyHit event at the actual impact frame.
- Remove or neutralize the Behavior Graph Attack damage if the script below owns damage; do not apply damage twice.
- Full-body EnemyAttack locks AI movement during the action through BlocksMovement integration.

Behavior Graph after the change

Branch	Purpose
Selector	Choose chase or patrol
Chase task	Path toward the player; Repath service updates route
Patrol/Wander	Fallback when target is not detected
EnemyCombatAnimation script	Starts attack in range and applies event-timed damage

OWNERSHIP RULE

Choose one damage owner. This expansion uses the EnemyHit animation event in EnemyCombatAnimation, so the graph should not also deal Attack task damage.

Synchronize enemy damage to EnemyHit

The enemy script starts a full-body action when the player is close, observes a cooldown, and damages the player only when the event fires.

ENEMYCOMBATANIMATION CORE

```
class EnemyCombatAnimation : public engine::Script {
    float cooldown = 0.0f;
public:
    void OnUpdate(float dt) override {
        cooldown = std::max(0.0f, cooldown - dt);
        const auto player = FindObject("PlayerStart");
        auto* me = Transform();
        auto* target = FindTransform("PlayerStart");
        if (!me || !target || player == engine::ecs::kNull) return;

        const float range = GetFieldFloat("range", 1.8f);
        if (glm::length(target->position - me->position) <= range &&
            cooldown <= 0.0f && !IsAnimationActionPlaying()) {
            if (PlayAnimationProfile("EnemyAttack"))
                cooldown = GetFieldFloat("cooldown", 1.2f);
        }

        if (WasAnimationEvent("EnemyHit")) {
            if (auto* hp = TryGet<engine::Health>(player))
                hp->Damage(GetFieldFloat("damage", 10.0f));
        }
    }
};
```

Field	Starter	Tune
range	1.8	Match weapon reach
cooldown	1.2	Longer than spam, near clip length
damage	10	Ten hits from 100 HP

Connect effects to animation events

Events are the shared timing source for projectiles, impact sound, particles, camera shake, and UI feedback.

Event	Gameplay	Feedback
StaffRelease	Spawn fireball	Staff flash + cast audio
Projectile hit	Damage Enemy	Impact burst + hit audio
EnemyHit	Damage Player	Hit particles + camera shake
Enemy death	Disable combat / death pose	Death audio/effect
Player death	GameMode defeat	HUD message + defeat stinger

Practical setup

- Put a burst Particle System on FireballPrototype; keep emission bounded by lifetime.
- Use spatial audio on the fireball/impact and UI or music buses for global stingers.
- Call ShakeCamera only for local player damage; keep intensity and duration restrained.
- If an event also uses an Audio.* command in Animation Preview, do not duplicate the same sound in script.
- Use material emissive color and light carefully; many dynamic lights can dominate particle cost.

EVENT NAMES ARE API

StaffRelease and EnemyHit are case-sensitive contracts between authored clips and scripts. Rename them in both places together.

Diagnose animation and combat in order

Watch the state that owns the failing behavior rather than tuning several systems at once.

Symptom	Inspect first	Likely fix
Model faces down	Import basis/object transform	Correct orientation before graph tuning
Feet slide	Gameplay speed vs stride	Tune clip/state speed or movement speed
Cast restarts	Input/script call frequency	Use WasMouseButtonPressed + action-playing guard
Player still moves	Action mask	Use Full Body/empty mask
Projectile never appears	StaffRelease event/runtime console	Correct clip index/time/name
Projectile misses trigger	Collider/physics events	Trigger sizes and target collider
Enemy double damages	Graph Attack + script event	Choose one damage owner
Enemy attacks while moving	Profile mask/entity split	Full-body action on moving entity
Damage timing wrong	EnemyHit marker	Scrub and reposition event

- Animation Preview > Runtime: current clip/time/blend/action.
- Graph Parameters and Transition Debug: live locomotion state.
- Physics Status: trigger enter events for projectile and target.
- Script Debug: enabled class, registration, fields, errors.
- Behavior Graph debug: chase branch and nav path status.

Verify and ship the expansion

Run the complete collection loop with animation and combat before exporting.

- 1 **Player model stands upright and matches capsule.**
- 2 **Player Idle/Walk/Run transitions correctly.**
- 3 **StaffCast plays once, locks or layers as designed, then returns.**
- 4 **StaffRelease spawns exactly one projectile.**
- 5 **Projectile travels, triggers, damages once, and expires.**
- 6 **Enemy locomotion matches navigation speed.**
- 7 **EnemyAttack locks movement; EnemyHit damages once.**
- 8 **No graph/script double damage.**
- 9 **Health HUD and game-over still work.**
- 10 **Collecting all coins still produces victory.**
- 11 **Audio/particles remain bounded during repeated combat.**
- 12 **F7 export and F8 runtime validation pass.**
- 13 **Standalone victory and defeat playthroughs pass.**

REGRESSION GATE

The expansion is complete only when every original Coin Rush acceptance test still passes with the new animated assets and scripts.

Files, registration, and the next expansion

Item	Location/role
Scene	Scenes/CoinRush_Animated.scene
Runtime scene	F7-generated CoinRush_Animated.runtime.scene
StaffCombat.h	game/include/game/scripts/
FireballProjectile.h	game/include/game/scripts/
EnemyCombatAnimation.h	game/include/game/scripts/
Registration	game/src/GameModule.cpp
Character assets	Project Assets tree
Package content	PLAYER_GAME_DIR bundled under game/

- Register all three exact class names and rebuild editor/player.
- The code listings show class cores; add `#pragma once` plus `Script.h`, `GameplayComponents.h`, `glm`, and standard-library includes used by each complete header.
- Save script fields on their scene objects: projectile speed/damage/lifetime and enemy range/cooldown/damage.
- Keep `FireballPrototype` in the exported scene so `SpawnFromObject` can clone it.
- Update `player.cfg` to the animated runtime scene and rebuild the release package.

NEXT COMPANION

Expansion 02 can add a main menu, pause/settings UI, multiple scenes, scene transitions with `RequestSceneLoad`, checkpoints, and saved options.

Related guide

Use the separate 3DENGINE Animation Preview Panel Guide for deeper State Graph, transition, mask, root-motion, and runtime-diagnostics reference.